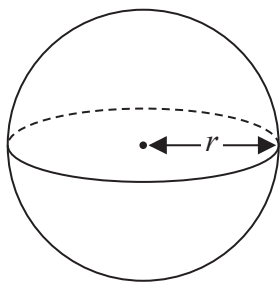
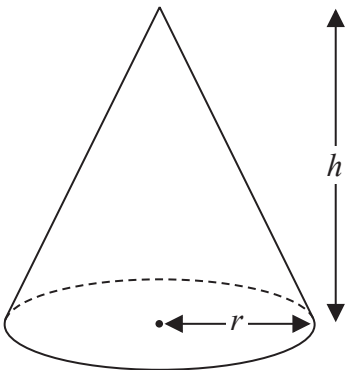


24 Here is a solid sphere and a solid cone.



Volume of sphere = $\frac{4}{3} \pi r^3$



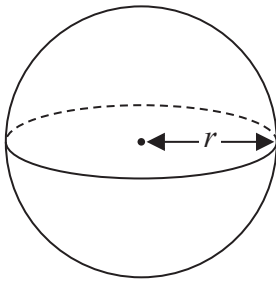
Volume of cone = $\frac{1}{3} \pi r^2 h$

All measurements are in cm.

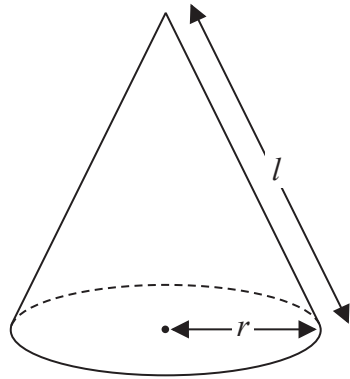
The volume of the sphere is equal to the volume of the cone.

- (a) Find $r:h$
Give your answer in its simplest form.

Here is a different solid sphere and a different solid cone.



$$\text{Surface area of sphere} = 4\pi r^2$$



$$\text{Curved area of cone} = \pi r l$$

All measurements are in cm.

The surface area of the sphere is equal to the **total** surface area of the cone.

(b) Find $r:h$

Give your answer in the form $1:\sqrt{n}$ where n is an integer.